

ISI UGA Mock 1: Complete Solutions

Problem 1

Solution

Instead of evaluating the integral directly, we look at how the sequence changes by subtracting consecutive terms:

$$a_n - a_{n-1} = \int_0^{\pi/2} \frac{\sin((2n+1)x) - \sin((2n-1)x)}{\sin x} dx.$$

Using the identity $\sin A - \sin B = 2 \sin\left(\frac{A-B}{2}\right) \cos\left(\frac{A+B}{2}\right)$, the numerator becomes $2 \sin(x) \cos(2nx)$. The $\sin x$ terms cancel perfectly:

$$a_n - a_{n-1} = \int_0^{\pi/2} 2 \cos(2nx) dx = \left[\frac{\sin(2nx)}{n} \right]_0^{\pi/2} = 0.$$

Since the difference is 0, the sequence never changes. Every term equals $a_0 = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$.
(Answer: C)

Quick approach

The options are constants. If a sequence of integrals evaluates to a constant, just calculate the easiest term ($n = 0$).

Remark

Telescoping sums and differences are a brilliant way to bypass terrifying limits.

Problem 2

Solution

We are solving $\sin(\cos x) = \cos(\sin x)$. The outputs of $\sin x$ and $\cos x$ are strictly between -1 and 1 , so we are plugging numbers in $[-1, 1]$ (in radians) into the outer functions.

For any $y \in [-1, 1]$, $\cos(y)$ is positive, with its lowest value being $\cos(1) \approx 0.54$. Meanwhile, the highest that $\sin(z)$ can reach for $z \in [-1, 1]$ is $\sin(1) \approx 0.84$. Because the cosine curve stays completely above the sine curve in the window from -1 to 1 , they never intersect. There are **0 solutions**. **(Answer: A)**

Quick approach

Draw a quick mental graph of $y = \sin(x)$ and $y = \cos(x)$ for small inputs between -1 and 1 . They don't cross.

Remark

Before wrestling with algebra, always check if the two sides of an equation even exist in the same numerical neighbourhood.

Problem 3

Solution

Let f be a continuous, non-decreasing curve from $(0,0)$ to $(1,1)$. The integral $\int_0^1 f(x) dx$ is the area between the curve and the x -axis. The integral $\int_0^1 f^{-1}(y) dy$ is the area between the same curve and the y -axis.

Drawing this shows that the two puzzle pieces snap together to fill the entire 1×1 square. The sum of their areas is simply $1 \times 1 = 1$. **(Answer: B)**

Quick approach

Pick the easiest function that fits: $f(x) = x$. Its inverse is $f^{-1}(y) = y$. Integrating both from 0 to 1 gives $\frac{1}{2} + \frac{1}{2} = 1$.

Problem 4

Solution

We need solutions to $x^2 + y^2 \equiv 1 \pmod{p}$. Draw a line from $(-1,0)$ through the circle with equation $y = t(x+1)$, where t is the slope. Substituting into the circle gives $x^2 + t^2(x+1)^2 = 1$, which factors as

$$(x+1)(x-1+t^2(x+1)) = 0.$$

This gives a unique point (x,y) for every slope t satisfying $1+t^2 \not\equiv 0 \pmod{p}$.

If $p \equiv 3 \pmod{4}$, there are no solutions to $t^2 \equiv -1$, leaving all p slopes valid. Adding the start point $(-1,0)$ gives $p+1$ points.

If $p \equiv 1 \pmod{4}$, there are two “bad” slopes where $t^2 \equiv -1$, leaving $p-2$ valid slopes. Adding $(-1,0)$ gives $p-1$ points.

Both cases are elegantly summarised by $p - (-1)^{(p-1)/2}$. **(Answer: C)**

Quick approach

Test $p = 3$. The squares mod 3 are 0 and 1. We need two squares summing to 1 ($1+0$ or $0+1$). This gives pairs $(1,0), (2,0), (0,1), (0,2)$, totalling 4. Among the options, only $3 - (-1)^1 = 4$ works.

Problem 5

Solution

Let $\alpha = \zeta + \zeta^2 + \zeta^4$ and $\beta = \zeta^3 + \zeta^5 + \zeta^6$. Expanding the product:

$$\alpha\beta = \zeta^4 + \zeta^6 + \zeta^7 + \zeta^5 + \zeta^7 + \zeta^8 + \zeta^7 + \zeta^9 + \zeta^{10}.$$

Since $\zeta = e^{2\pi i/7}$, we have $\zeta^7 = 1$. Reducing larger exponents ($\zeta^8 = \zeta$, $\zeta^9 = \zeta^2$, etc.) gives

$$\alpha\beta = 3(1) + \zeta + \zeta^2 + \zeta^3 + \zeta^4 + \zeta^5 + \zeta^6.$$

The sum of all 7th roots of unity is $1 + \zeta + \cdots + \zeta^6 = 0$, so the ζ terms sum to -1 . The result is $3 - 1 = 2$. **(Answer: B)**

Quick approach

Recognising that the exponents $\{1, 2, 4\}$ and $\{3, 5, 6\}$ are the quadratic residues and non-residues modulo 7 makes this expansion predictable.

Problem 6

Solution

Let the roots be z_1, z_2, z_3 . Since they lie on the unit circle, $|z_1| = |z_2| = |z_3| = 1$. The polynomial is $P(x) = x^3 - ax^2 + bx - c$.

The constant term is $c = z_1 z_2 z_3$. Since $|c| = 1$ and c is an integer, $c \in \{1, -1\}$. The coefficient $a = z_1 + z_2 + z_3$ satisfies $|a| \leq 3$ by the triangle inequality, so $a \in \{-3, \dots, 3\}$.

For the coefficient $b = z_1 z_2 + z_2 z_3 + z_3 z_1$, observe

$$b = z_1 z_2 z_3 \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right).$$

On the unit circle, $1/z = \bar{z}$, so $b = c\bar{a}$. Since a is real, $b = ac$.

We need only check polynomials of the form $x^3 - ax^2 + acx - c = 0$:

If $c = 1$: $P(x) = (x-1)(x^2 + (1-a)x + 1)$. For the quadratic roots to lie on the unit circle, $(1-a)^2 - 4 \leq 0$, giving $-1 \leq a \leq 3$. That is 5 valid polynomials.

If $c = -1$: $P(x) = (x+1)(x^2 - (1+a)x + 1)$. The condition $(1+a)^2 - 4 \leq 0$ gives $-3 \leq a \leq 1$. Again 5 valid polynomials.

Total = 10. **(Answer: D)**

Remark

Using the property $1/z = \bar{z}$ completely avoids Kronecker's Theorem — a powerful, elementary algebraic shortcut.

Problem 7

Solution

Let $q(x) = p(x) + p'(x) + p''(x) + \cdots$ (a finite sum, since p is a polynomial). Notice that

$$q(x) - q'(x) = p(x) \geq 0.$$

Since $p(x) \geq 0$ everywhere, its highest power is even and its leading coefficient is positive. The polynomial q shares this same highest power, so $q(x) \rightarrow +\infty$ on both ends and must attain a global minimum at some point x_0 .

At this valley, the slope is zero: $q'(x_0) = 0$. Substituting into the relation gives $q(x_0) = p(x_0) \geq 0$. Since the lowest point of q is non-negative, $q(x) \geq 0$ for every real x . **(Answer: A)**

Quick approach

Take $p(x) = x^2 + 1$. Its derivatives are $2x$ and 2 . The sum is $x^2 + 2x + 3 = (x + 1)^2 + 2$, which is clearly always ≥ 2 .

Problem 8

Solution

We want to maximise the cyclic sum $a^2b + b^2c + c^2a + abc$. We may assume b lies between a and c , so $(b - a)(b - c) \leq 0$. Multiplying by a gives $a^2c + ab^2 \leq a^2b + abc$. Substituting into the expression:

$$a^2b + b^2c + c^2a + abc \leq b(a + c)^2.$$

Now apply AM–GM to b , $\frac{a+c}{2}$, $\frac{a+c}{2}$:

$$b(a + c)^2 = 4 \cdot b \cdot \frac{a+c}{2} \cdot \frac{a+c}{2} \leq 4 \left(\frac{b + \frac{a+c}{2} + \frac{a+c}{2}}{3} \right)^3 = 4 \left(\frac{a + b + c}{3} \right)^3 = 4.$$

(Answer: B)

Quick approach

Test a balanced asymmetric distribution: $a = 2$, $b = 1$, $c = 0$. The expression gives $4(1) + 0 + 0 + 0 = 4$.

Problem 9

Solution

A 5-digit number with strictly increasing digits is formed by picking 5 distinct digits from $\{1, 2, \dots, 9\}$ (zero cannot lead). The sum of all 9 digits is 45, which is divisible by 3. Choosing 5 digits whose sum is divisible by 3 is the same as leaving behind 4 digits whose sum is divisible by 3.

Partition the digits by residue mod 3: $A = \{3, 6, 9\}$, $B = \{1, 4, 7\}$, $C = \{2, 5, 8\}$. We need (n_A, n_B, n_C) summing to 5 items with residue sum $\equiv 0 \pmod{3}$. The valid distributions are $(3, 1, 1)$, $(2, 3, 0)$, $(2, 0, 3)$, and $(1, 2, 2)$:

$$\binom{3}{3} \binom{3}{1} \binom{3}{1} + \binom{3}{2} \binom{3}{3} \binom{3}{0} + \binom{3}{2} \binom{3}{0} \binom{3}{3} + \binom{3}{1} \binom{3}{2} \binom{3}{2} = 9 + 3 + 3 + 27 = 42.$$

(Answer: A)

Problem 10

Solution

For t close to 1, $\ln t$ is bounded by $\frac{t-1}{t} < \ln t < t-1$. Inverting:

$$\frac{1}{t-1} < \frac{1}{\ln t} < 1 + \frac{1}{t-1}.$$

Integrating the lower bound from x to x^2 gives $\ln(x^2-1) - \ln(x-1) = \ln(x+1)$. As $x \rightarrow 1$, this approaches $\ln 2$.

Integrating the upper bound gives $(x^2-x) + \ln(x+1)$. As $x \rightarrow 1$, the (x^2-x) part vanishes, leaving $\ln 2$.

By the Squeeze Theorem, the integral equals $\ln 2$. **(Answer: B)**

Remark

Bounding tricky functions with rational polynomials before squeezing is a fantastic, rigorous technique.

Problem 11

Solution

For any non-right triangle, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. For acute triangles, all tangents are positive. Applying AM-GM to the sum: $\frac{\tan A + \tan B + \tan C}{3} \geq (\tan A \tan B \tan C)^{1/3}$.

Writing $P = \tan A \tan B \tan C$, this becomes $P/3 \geq P^{1/3}$, which gives $P^{2/3} \geq 3$, so $P \geq 3\sqrt{3}$. **(Answer: B)**

Quick approach

Extrema in triangle problems almost always occur at the equilateral triangle. For $A = B = C = 60^\circ$, the product is $(\sqrt{3})^3 = 3\sqrt{3}$.

Problem 12

Solution

Consider $\triangle ABD$ and $\triangle AEC$. The angle bisector gives $\angle BAD = \angle EAC$. Also, $\angle ABD$ and $\angle AEC$ subtend the same arc AC on the circumcircle, so they are equal. With two equal angles, $\triangle ABD \sim \triangle AEC$.

Proportional sides give $\frac{AB}{AE} = \frac{AD}{AC}$, so $AB \cdot AC = AD \cdot AE$. With $AB = 6$ and $AC = 4$, the product is 24. **(Answer: B)**

Problem 13

Solution

Given $A^2 = -I$, take determinants: $(\det A)^2 = \det(-I) = (-1)^n$. Since A is real, $(\det A)^2 \geq 0$, forcing $(-1)^n = 1$, so n must be even. (Confirms B.)

Since n is even, $(\det A)^2 = 1$. The eigenvalues satisfy $\lambda^2 = -1$, giving $\lambda = \pm i$. Since A is real, complex eigenvalues appear in conjugate pairs: $n/2$ copies of i and $n/2$ copies of $-i$. The determinant is

$$i^{n/2} \cdot (-i)^{n/2} = (i \cdot (-i))^{n/2} = 1^{n/2} = 1.$$

(Confirms C.) **(Answer: D)**

Problem 14

Solution

The recurrence is $a_{n+1} = \frac{1}{2}(a_n + \frac{2}{a_n})$. By AM–GM, $\frac{1}{2}(a_n + \frac{2}{a_n}) \geq \sqrt{2}$, so every term is $\geq \sqrt{2}$.

Subtracting a_n from both sides shows the sequence is decreasing. A decreasing sequence bounded below converges to a limit L . Substituting L into the recurrence gives $L = \frac{L^2+2}{2L}$, so $L^2 = 2$ and $L = \sqrt{2}$. **(Answer: B)**

Problem 15

Solution

Given $f(x+y) + f(x-y) = 2f(x)f(y)$, set $y = 0$: this gives $2f(x) = 2f(x)f(0)$. Since f is not identically zero, $f(0) = 1$. The only continuous functions satisfying this functional equation with $f(0) = 1$ are $\cos(cx)$ and $\cosh(cx)$. **(Answer: A)**

Quick approach

This directly mirrors the classic identity $\cos(x+y) + \cos(x-y) = 2\cos x \cos y$.

Problem 16

Solution

We need $n+10 \mid n^3+100$. Factor:

$$n^3+100 = (n^3+10^3) - 900 = (n+10)(n^2-10n+100) - 900.$$

The first term is divisible by $n+10$, so $n+10$ must divide 900. To maximise n , take the largest factor of 900, which is 900 itself: $n+10 = 900$, giving $n = 890$. **(Answer: A)**

Quick approach

By the Remainder Theorem, set $n = -10$ in n^3+100 to get $(-10)^3+100 = -900$. The answer reduces to finding the largest divisor of 900.

Problem 17

Solution

Complex numbers on the unit circle have magnitude 1. If $z_1 + z_2 + z_3 = 0$, their centroid is at the origin. For a triangle inscribed in a circle, if the centroid and the circumcentre coincide, the triangle must be equilateral.

An equilateral triangle inscribed in a circle of radius $R = 1$ has side length $\sqrt{3}$. The sum of the squares of the three sides is $3 \times (\sqrt{3})^2 = 9$. **(Answer: B)**

Problem 18

Solution

Define $h(x) = f(x) - \frac{M}{2}x(x-1)$. Its second derivative is $h''(x) = f''(x) - M \leq 0$, so h is concave. Since $h(0) = 0$ and $h(1) = 0$, the concave dome stays above the x -axis: $h(x) \geq 0$, giving $f(x) \geq \frac{M}{2}x(x-1)$.

A symmetric argument traps f from above, yielding $|f(x)| \leq \frac{M}{2}x(1-x)$. The maximum of $x(1-x)$ on $[0, 1]$ is $\frac{1}{4}$ (at $x = \frac{1}{2}$), so $|f(x)| \leq M/8$. **(Answer: D)**

Problem 19

Solution

The only way two integers sum to an odd number is for one to be even and one to be odd. In $\{1, 2, \dots, 2n\}$, there are exactly n odd and n even numbers.

Line up the n odd numbers. The n even numbers can be paired with them in $n!$ ways. **(Answer: A)**

Problem 20

Solution

We track factors of 3 using $x^3 - 1 = (x-1)(x^2 + x + 1)$. If $x = 3k + 1$, the second factor becomes $(3k+1)^2 + (3k+1) + 1 = 9k^2 + 9k + 3 = 3(3k^2 + 3k + 1)$. This is divisible by 3 but not by 9.

So every time we cube the exponent, we multiply by a bracket that adds exactly one factor of 3. We start with $10^1 - 1 = 9$, which has two factors of 3. To reach $10^{3^{10}}$, we cube the exponent 10 times, adding 10 more factors. Total = $2 + 10 = 12$. **(Answer: C)**

Remark

This builds the exact intuition behind the Lifting the Exponent (LTE) lemma without memorising any formulas.

Problem 21

Solution

Apply Cauchy-Schwarz with $x_i = 1$ and $y_i = \sqrt{4a_i + 1}$:

$$(\sqrt{4a+1} + \sqrt{4b+1} + \sqrt{4c+1})^2 \leq 3(4a+1 + 4b+1 + 4c+1) = 3(4(a+b+c) + 3).$$

Since $a + b + c = 1$, the right side is $3 \times 7 = 21$. The sum is at most $\sqrt{21}$. **(Answer: B)**

Quick approach

Symmetric functions usually hit their max when all variables are equal. Setting $a = b = c = \frac{1}{3}$ gives $3\sqrt{4/3+1} = \sqrt{21}$.

Problem 22

Solution

Let the circle centres be O_1, O_2 . Drop perpendiculars from the centres to AB ; their feet X, Y are the midpoints of PA and PB (perpendiculars from the centre bisect chords). The length XY is the projection of O_1O_2 onto the rotating line.

The midpoint M of AB satisfies $\vec{PM} = \vec{PX} + \vec{PY}$, and forms a constant angle with P . Geometrically, its locus traces a circle passing through P . **(Answer: B)**

Problem 23

Solution

Given $AB - BA = A$, take the trace: $\text{tr}(AB - BA) = 0$, so $\text{tr}(A) = 0$.

If A were invertible, multiplying on the left by A^{-1} gives $B - A^{-1}BA = I$, i.e. $B = A^{-1}BA + I$. But $A^{-1}BA$ is similar to B and must share its eigenvalues, while adding I shifts every eigenvalue by 1. A matrix cannot have the same eigenvalues as itself shifted by 1. So A is not invertible, meaning $\det A = 0$. **(Answer: B)**

Problem 24

Solution

The sum $S_n = \frac{1}{n} \sum_{k=1}^n \frac{1}{1+k/n}$ is the right Riemann sum for $\int_0^1 \frac{dx}{1+x} = \ln 2$. Because $f(x) = 1/(1+x)$ is decreasing, the right-hand rectangles sit slightly below the curve.

The error looks like tiny triangles on top of the rectangles. Their total area telescopes:

$$\text{Error} \approx \frac{1}{2n}(f(0) - f(1)).$$

Since S_n underestimates, $S_n - \ln 2 \approx \frac{1}{2n}(f(1) - f(0))$. Multiplying by n and taking the limit:

$$\lim_{n \rightarrow \infty} n(S_n - \ln 2) = \frac{1}{2}(f(1) - f(0)) = \frac{1}{2}\left(\frac{1}{2} - 1\right) = -\frac{1}{4}.$$

(Answer: B)

Remark

Drawing the sliding triangles proves the approximation error using simple geometry, bypassing heavy calculus formulas.

Problem 25

Solution

The fractional part function $\{y\}$ is continuous everywhere except when y is an integer. The term $\{nx\}$ jumps whenever nx is an integer, i.e. when $x = k/n$ for some integer k . Since k and n are integers, x must be rational.

Because the infinite series converges smoothly, the sum inherits these exact breaking points. It fails to be continuous precisely on all rational numbers $\mathbb{Q}$. (**Answer: B**)

Problem 26

Solution

The polynomial gives an odd output for $p(0), p(1), \dots, p(n)$. Suppose there is an integer root r , so $p(r) = 0$. Since 0 is even, $p(r)$ is even.

If r is even, then $p(r)$ has the same parity as $p(0)$. But $p(0)$ is odd — contradiction. If r is odd, then $p(r)$ has the same parity as $p(1)$. But $p(1)$ is odd — contradiction again.

The polynomial can never output an even number at an integer, so it can never output 0. (**Answer: A**)

Problem 27

Solution

Let a_n be the number of ways to write n as an ordered sum of odd positive integers. Consider the last summand.

If the sum ends in 1, removing it leaves a valid sum for $n - 1$: this gives a_{n-1} possibilities. If the sum ends in an odd number $k \geq 3$, subtracting 2 from it gives a valid sum for $n - 2$ ending in $k - 2 \geq 1$: this maps perfectly to a_{n-2} possibilities.

Combining, $a_n = a_{n-1} + a_{n-2}$. Since $a_1 = 1$ and $a_2 = 1$, this is exactly the Fibonacci sequence. So $a_{100} = F_{100}$. (**Answer: B**)

Problem 28

Solution

The minimum number of distinct distances among n points on a line occurs when the points are equally spaced (e.g. $1, 2, 3, \dots, n$). In that case, the distances are $1, 2, \dots, n - 1$, giving $n - 1$ distinct values. One cannot achieve fewer than $n - 1$ distinct distances with n distinct collinear points. (**Answer: C**)

Problem 29

Solution

This is the famous Takagi curve (blancmange curve). The weights 2^{-n} make the wave heights shrink fast enough that the function stays continuous everywhere. However, the factor $2^n \pi x$ inside the sine makes the oscillations infinitely rapid.

Because the function wiggles so violently at every microscopic scale, a stable tangent line can never be drawn. It is differentiable at exactly 0 points. **(Answer: A)**

Problem 30

Solution

We have $PA = 3$, $PB = 4$, $PC = 5$. Rotate $\triangle APB$ around point B by exactly 60° to form a new triangle $\triangle CQB$.

Because $BP = BQ = 4$ and the rotation angle is 60° , $\triangle PBQ$ is equilateral, so $PQ = 4$. By the rotation, $CQ = AP = 3$.

Now consider $\triangle PQC$: its sides are 4, 3, 5. This is a Pythagorean triple, so $\angle PQC = 90^\circ$.

In the larger $\triangle BQC$, we have $BQ = 4$, $QC = 3$, and $\angle BQC = 60^\circ + 90^\circ = 150^\circ$. By the law of cosines:

$$BC^2 = 4^2 + 3^2 - 2(4)(3)\cos 150^\circ = 25 + 12\sqrt{3}.$$

So $BC = \sqrt{25 + 12\sqrt{3}}$. **(Answer: A)**